



SAMRAT POLY RESINS

Manufacturer of Coating & Composite Resins · Doraha, Ludhiana, Punjab, India

TECHNICAL DATA SHEET

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ISO GELCOAT

Isophthalic Polyester Gelcoat · Mould-Surface Coat for FRP Mouldings

PRODUCT DESCRIPTION

ISO Gelcoat is an isophthalic polyester gelcoat manufactured by Samrat Poly Resins. A gelcoat is the outermost layer of an FRP moulding, applied to the mould face before any reinforcement goes in, and it gives the finished part its visible surface and its first line of protection. Isophthalic gelcoats are the grade normally specified where that surface has to hold gloss and colour through weathering, water contact or outdoor service — marine, sanitaryware and transport work being the common examples. It is a separate grade from the general-purpose GP Gelcoat Resin.

KEY FEATURES

- Isophthalic gelcoat chemistry rather than a general-purpose polyester gelcoat
- Forms the visible surface and first line of protection of the moulded part
- Applied to the mould face before any reinforcement
- Pre-accelerated, thixotropic, for brush and spray application
- MEKP catalysed, for hand lay-up and spray-up FRP moulding

RECOMMENDED APPLICATIONS

- Mould-face surface coat for FRP mouldings
- Sanitaryware and transport panel moulding
- Exterior panel moulding
- Parts whose surface must hold gloss and colour in service

PRODUCT SPECIFICATION

Working draft — the test conditions marked below are pending confirmation.

IDENTITY AND SUPPLY		
Resin Type	Isophthalic Polyester Gelcoat	supplied by Samrat Poly Resins
Backbone	Isophthalic unsaturated polyester	supplied by Samrat Poly Resins
Grade	ISO Gelcoat	supplied by Samrat Poly Resins
Colour / Appearance	Creamy off-white to pale ivory thixotropic gelcoat	supplied by Samrat Poly Resins
Supplied Form	Pigmented / pigmentable thixotropic liquid	supplied by Samrat Poly Resins
Application Form	Brush and spray application	supplied by Samrat Poly Resins
Acceleration Status	Pre-accelerated	supplied by Samrat Poly Resins
Country of Origin	India	supplied by Samrat Poly Resins
LIQUID GELCOAT PROPERTIES		
Viscosity — low shear	40,000 cPs	at 25 °C; spindle and speed — to be confirmed
Viscosity — high shear	1,500 cPs	at 25 °C; spindle and speed — to be confirmed
Thixotropic Index	26, as supplied	the quotient of the two viscosities above is 26.7 — confirm the speed pair the index is calculated on
Gel Time	8 minutes	at 25 °C; catalyst level used in this determination — to be confirmed
Specific Gravity	1.15 g/cm ³	at 25 °C
Styrene Content	32% by weight	determination method — to be confirmed
Flash Point	32 °C	closed cup
APPLICATION		
Process	Hand lay-up and spray-up FRP moulding	supplied by Samrat Poly Resins
Catalyst and Dosage	MEKP at 2.0% by weight of gelcoat	supplied by Samrat Poly Resins
Wet Film Thickness	700 µm target; 600–800 µm recommended range	supplied by Samrat Poly Resins
Number of Coats	One complete gelcoat layer	supplied by Samrat Poly Resins
Application Temperature	15–25 °C	supplied by Samrat Poly Resins
Back-up / Laminating Time	30–45 minutes	supplied by Samrat Poly Resins
CURED GELCOAT PROPERTIES		
Barcol Hardness	42	ASTM D2583; cure condition at test — to be confirmed
Gloss	90 GU at 60°	ISO 2813 or ASTM D523 — to be confirmed
Water Resistance	Good to very good	qualitative rating

Water Absorption	Approximately 18 mg after 24 h at 23 °C	<i>specimen size and mass, and test method — a milligram result cannot be read without them — to be confirmed</i>
WEATHERING, AS SUPPLIED		
Test Method	ISO 4892-2 xenon arc	<i>exposure cycle, irradiance, filter and black-standard temperature — to be confirmed</i>
Alternative Method	ASTM G155 xenon arc	<i>named as a comparable alternative method; not a second exposure that was performed</i>
Exposure Duration	1,000 hours	<i>supplied by Samrat Poly Resins</i>
Colour Change	ΔE 3–5	<i>colour measurement standard — to be confirmed</i>
Gloss Retention	80%	<i>gloss geometry and measurement standard — to be confirmed</i>
STORAGE		
Shelf Life	3 months	<i>in original tightly closed containers at the storage conditions below</i>
Storage Temperature	15–25 °C, maximum 25 °C	<i>supplied by Samrat Poly Resins</i>
Storage Conditions	Original tightly closed containers in a cool, dry, well-ventilated area. Protect from sunlight, heat, sparks, ignition sources and moisture.	<i>supplied by Samrat Poly Resins</i>

Values in this table are supplied by Samrat Poly Resins for this grade. They are typical values for the product as manufactured, not guaranteed limits. Conditions marked “to be confirmed” are test conditions not yet stated; they are shown as outstanding rather than assumed. The thixotropic index is printed as supplied; see the note against that row.

GRADE SELECTION

For standard-duty FRP mould work where the part is not in sustained water contact or outdoor service, GP Gelcoat Resin is the general-purpose route. A gelcoat is a surface coating and not a laminating resin: the structural laminate behind it is built with a separate resin such as GP Clear Resin, or ISO Polyester Resin for chemical and water service. Share the finished part, its service condition and the required finish with your enquiry.

SAFETY & HANDLING

Handle with appropriate personal protective equipment and adequate ventilation. Polyester gelcoats contain styrene, which is flammable. Keep away from heat, sparks and open flame. **Never mix the MEKP catalyst directly with a cobalt accelerator. Add each separately — direct contact between the two can react violently.** Refer to the applicable Safety Data Sheet (SDS) before handling.

Disclaimer: The information above is based on data published by the manufacturer at the time of issue and is believed accurate. Values shown are typical and are not to be interpreted as guaranteed specifications; properties may vary within normal manufacturing tolerances between batches. Users are responsible for determining suitability for their intended application and should confirm current specifications directly with Samrat Poly Resins prior to use.